

WIRELESS NETWORKING

Day – 9 Fireball basic theory.

Task –1 Opening the defender fireball.

- Steps to Block Spotify Outbound Traffic :-
- 1.Open Windows Defender Firewall:30 sec.
- Press Windows Key + R to open the Run box, type wf.msc, and press Enter to open Windows Defender Firewall with Advanced Security.
- 2.Start New Outbound Rule Wizard:30 sec.
- In the left pane, click on Outbound Rules. Then, in the right pane under Actions, click New Rule....
- 3.Select Program Type and Path:1 min.
- Select Program as the Rule Type and click Next. Choose This program path: and click Browse.... Locate the Spotify executable file (typically found at %APPDATA%\Spotify\Spotify.exe), select it, and click Next.
- 4.Set Action to Block:30 sec.
- Select Block the connection and click Next. Keep all profile boxes checked (Domain, Private, and Public) and click Next.
- 5.Name and Apply:30 sec.
- Name the rule Block Spotify App and click Finish.

Task – 2 Inbound rule in defender firewall.

- Step-by-Step Setup
- 1.Open Firewall Advanced Security:30 sec.
- Press Windows Key + R, type wf.msc, and press Enter.
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- 2.Start New Inbound Rule Wizard:30 sec.
- Click Inbound Rules in the left panel, then click New Rule... in the right panel under *Actions*.
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- 3.Select Custom Rule Type:30 sec.

- Select Custom and click Next.
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- 4. Select Chrome Executable: 1 min.
- Select This program path:, click Browse..., navigate to C:\Program Files\Google\Chrome\Application\chrome.exe, and click Next.
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- 5. Specify Protocol and Port 443: 1 min.
- Set Protocol type: to TCP.
- Set Local port: to Specific Ports.
- Type 443 in the port field and click Next.
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- 6. Set Action and Name Rule: 30 sec.
- Leave scope settings as Any IP address and click Next.
- Select Allow the connection and click Next.
- Check all profile boxes (Domain, Private, Public) and click Next.
- Name the rule Allow Chrome Inbound HTTPS 443 and click Finish.
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- Verify :- In the Inbound Rules list, locate Allow Chrome Inbound HTTPS 443.
- Right-click the rule and choose Properties.
- Check the Programs and Services tab (verifying chrome.exe) and the Protocols and Ports tab (verifying TCP port 443) to confirm your specifications are active.

Task – 3 Blocking the traffic from the specific IP address.

- Step-by-Step Setup
- 1. Open Firewall Advanced Security: 30 sec.
- Press Windows Key + R, type wf.msc, and press Enter.
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- 2. Start New Inbound Rule Wizard: 30 sec.
- Click Inbound Rules in the left panel, then click New Rule... in the right panel under Actions.
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- 3. Select Custom Rule Type: 30 sec.
- Select Custom and click Next.
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- 4. Select All Programs & Protocols: 30 sec.
- Leave program selection set to All programs and click Next.

- Leave protocol selection set to Any and click Next.
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- 5.Specify Remote IP Address in Scope:1 min.
- Under Which remote IP addresses does this rule apply to?, select These IP addresses:.
- Click Add..., select This IP address or subnet:, type 123.45.67.89, click OK, and then click Next.
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- 6.Set Action, Profile, and Name Rule:30 sec.
- Select Block the connection and click Next.
- Keep all profiles checked (Domain, Private, Public) and click Next.
- Name the rule Block IP 123.45.67.89 Inbound and click Finish.
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- 7.Repeat for Outbound Rule (Optional):1 min.
- To block outbound traffic to that IP as well, repeat the process under Outbound Rules, setting the remote IP to 123.45.67.89 and selecting Block the connection.

Task – 4 Compare firewall hardware.

- Comparison: Hardware vs. Software Firewalls :-

Feature / Difference	Hardware Firewall	Software Firewall
Location & Deployment	Positioned as a dedicated physical appliance at the perimeter (between the internal network and the Internet).	Installed directly on individual operating systems (e.g., Windows, macOS, Linux).
Performance Impact	Processes traffic using dedicated onboard hardware, causing zero CPU overhead on host devices.	Consumes host system resources (CPU and RAM) to analyze network packets.
Protection Scope	Protects the entire network and all connected devices simultaneously.	Protects only the single device (host) on which it is installed.

- Use Cases :-
- Hardware Firewall Use Case: A corporate headquarters protecting 500+ employee workstations, internal servers, and confidential data from external internet threats at the entry point of the network.
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- Software Firewall Use Case: A remote worker using a laptop on public coffee shop Wi-Fi, using Windows Defender Firewall to block malicious incoming traffic targeting their specific machine.
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